

Organizing Data Using Frequency Distribution Tables

Jonathan R. Bacolod

A frequency distribution table is used to organize raw data into a meaningful form.

Frequency

Frequency tells how many times a value appears in a data set.

Example: If 5 appears 3 times, its frequency is 3.

Tally Mark

A **tally mark** is used to keep track of the number of occurrences of each value.

Value	Tally	Frequency
5	IIII	4

Categories, Scores, or Data Values

The values being counted are called **categories, scores, or data values**.

These are the numbers or items listed in the leftmost column of a frequency table.

Why Organize Data?

Organizing data into a frequency table makes it easier to:

- ▶ Compare values
- ▶ Analyze patterns

Total Frequency

The sum of all frequencies is called the **total frequency**, which is equal to the number of data items.

$$\text{Total Frequency} = \sum (\text{individual frequencies})$$

Why This Matters

Frequency distribution tables are commonly used before constructing graphs such as bar graphs and pie graphs.

Mnemonic: *Tally it, Table it, THEN graph it!*

Steps to Build a Frequency Table

1. List all the different values (categories).
2. Tally each occurrence in the data set.
3. Count the tallies to get the frequency.
4. Add all frequencies to check the total.

Examples

Example 1

Data set:

3, 2, 3, 4, 2, 3, 4, 2, 3

Value	Tally	Frequency
2	III	3
3	IIII	4
4	II	2

Example 1 (continued)

$$\text{Total Frequency} = 3 + 4 + 2 = 9$$

This matches the 9 data values we started with,
so the table is correct.

Example 2

Data set (number of siblings):

1, 0, 1, 2, 0, 1, 2, 1, 0, 2

Value	Tally	Frequency
0	III	3
1	IIII	4
2	III	3

Example 2 (continued)

$$\text{Total Frequency} = 3 + 4 + 3 = 10$$

There were 10 students surveyed, so the total frequency checks out.

Practice Exercises

Practice 1

Organize the following data into a frequency distribution table.

5, 6, 5, 7, 8, 6, 5, 7, 6, 8, 5, 7, 6, 5, 8

Practice 2

Create a frequency distribution table for these daily study hours.

1, 2, 1, 3, 2, 1, 4, 2, 3, 1, 2, 4, 3, 2, 1

Practice 3

Make a frequency distribution table for the following numbers.

10, 11, 10, 12, 13, 11, 10, 12, 11, 13, 10, 11

Practice 4

Construct a frequency distribution table.

15, 16, 15, 17, 18, 16, 15, 17, 16, 18, 15, 17

Practice 5

Create a frequency distribution table.

4, 5, 6, 4, 5, 4, 7, 6, 5, 4, 6, 7, 5, 4, 6

Activities

Activity 1

Construct a frequency distribution table for the following favorite numbers.

1, 2, 3, 1, 4, 3, 2, 1, 4, 3, 1, 2, 3, 4, 1

Activity 2

Organize the following shoe sizes into a frequency distribution table.

37, 38, 39, 38, 37, 40, 39, 38, 37, 39, 40, 38

Activity 3

Organize the following ages into a frequency distribution table.

11, 12, 11, 13, 12, 11, 14, 13, 12, 11, 14, 13

Activity 4

Organize the following data.

24, 25, 26, 24, 27, 26, 25, 24, 27, 26, 24, 25

Activity 5

Organize the following test scores into a frequency distribution table.

65, 70, 75, 70, 65, 80, 75, 70, 65, 75, 80, 70

Thank you!